

Theory and implementation of reusable real-time TDDFT in Aion

Bare length and velocity gauges, covariant P0/P0+E1 dynamics,
currents, energies, connection-aware propagation, and spectroscopy

Aion project

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Abstract

This note fixes the physical definitions, sign and orientation conventions, equations of motion, observables, numerical algorithms, and software boundaries for the proposed Aion 0.2 real-time time-dependent density-functional theory (RT-TDDFT) implementation. It treats four formulations on one common finite-basis electronic reference: bare length gauge, bare velocity gauge, covariant P0, and covariant P0+E1. Particular attention is paid to quantities that cannot be transferred uncritically between formulations: the primary current, the mechanical-current diagnostic, matter and generator energies, source work, and the evolving-metric connection. The production propagation algorithm is self-consistent exponential midpoint (SCEM), with a connection-aware transport for the covariant formulations and no time-dependent Löwdin frame. The final sections map these definitions to a reusable API, storage schema, and validation hierarchy. This is primarily a normative design document. Its package, configuration, identity, schema, status, and API contracts were implemented in WP1. WP2 implements the common prepared electronic reference, backend workspaces, AO operators, and compiled electromagnetic source. WP3 implements all four formulations and their instantaneous dipoles, currents, energy ledgers, analytic rates, and observable definitions. WP4 implements the common nonlinear SCEM engine, direct fixed-metric propagation, and connection-aware covariant transport on both CPU and GPU. WP5 implements exact discrete-event execution, scheduled observers, interval power and work, transactional trajectories, checkpoint/restart, monitoring, export, and comparison manifests on both CPU and GPU. WP6 implements structured Casida response, polarized resonance selection, real-time kick spectra, dipole/current frequency-domain cross-checks, and immutable spectrum persistence; its kick propagation is qualified on the physical GPU.

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1 Document status and scope

1.1 Meaning of status labels

Four kinds of statements are distinguished throughout this note.

Established theory

Standard RT-TDDFT, minimal coupling, gauge transformation, evolving-Hilbert- space geometry, and rational approximation results supported by the cited literature.

Normative Aion 0.2

An agreed definition or software/numerical contract that the refactor must implement.

Draft-validated Physics or algorithms exercised in the present research tree, but not yet available through the clean 0.2 API or fully covered by its tests.

Planned/unvalidated

Agreed work whose complete reusable implementation and validation remain to be done.

Status — Normative Aion 0.2. When this note uses *must* or gives an unqualified stored name, it specifies the intended 0.2 contract. The implementation plan in `docs/refactor_implementation_plan.md` controls the order and acceptance gates. That plan is deliberately mutable; changes to the physics conventions require an explicit reviewed decision.

1.2 Validated physical domain

The first release targets finite, fixed-nucleus, all-electron molecules in restricted spin-summed Kohn–Sham DFT, with pure LDA or GGA functionals and prescribed spatially uniform electric fields. Atomic units are used internally and the numerical precision is double real/complex precision on CPU and GPU. The nuclear attraction is local.

The following are explicitly outside the first physics stack:

- pseudopotentials, effective-core potentials, and nonlocal ionic operators;
- hybrid and exact-exchange current corrections;
- unrestricted, noncollinear, or relativistic spin dynamics;
- moving nuclei and their basis-velocity connection;
- periodic boundary conditions;
- Maxwell backreaction and adaptive time grids.

These cases must be rejected rather than silently treated with formulas valid only for local all-electron Hamiltonians. The source and observable interfaces are nevertheless designed so that spatially varying fields, aloof relativistic projectiles, EELS, and possibly periodic velocity gauge can be added later without redefining the present uniform-field theory. They are extension seams, not present work packages.

2 Notation and conventions

2.1 Atomic basis and density

Let $\{|\phi_\mu(t)\rangle\}$ be a generally time-dependent nonorthogonal atomic-orbital (AO) basis. The overlap and occupied orbitals are

$$S_{\mu\nu}(t) = \langle\phi_\mu(t)|\phi_\nu(t)\rangle, \quad |\psi_i(t)\rangle = \sum_\mu |\phi_\mu(t)\rangle C_{\mu i}(t). \quad (1)$$

With occupations collected in f , the canonical AO density convention is

$$P = CfC^\dagger, \quad N = \text{Re Tr}(PS), \quad \langle O \rangle = \text{Re Tr}(PO). \quad (2)$$

The occupations contain the complete spin multiplicity. No contraction may insert an implicit closed-shell factor of two. A different covariant/dual or orthonormal representation is a different type and requires an explicit conversion.

Electronic charge and mass are kept symbolically as q and m ; electrons have $q = -1$, $m = 1$ in atomic units. We retain $\hbar$ in derivations to expose phases and signs, although $\hbar = 1$ in the implementation. The reduced vector potential is

$$\mathbf{a} = \frac{\mathbf{A}}{c}, \quad (3)$$

and the full stored name is `vector_potential_reduced`. This avoids an otherwise persistent ambiguity about factors of c .

2.2 Coordinates, origin, and molecular observables

Input coordinates are preserved exactly: Aion must never silently recenter a molecule. An explicit electromagnetic origin $\mathbf{O}$, defaulting to the input-coordinate origin, is stored. Electronic, fixed-nuclear, and total dipoles use this same origin:

$$\boldsymbol{\mu}_e = q \text{Re Tr}[P(\mathbf{r} - \mathbf{O}S)], \quad (4)$$

$$\boldsymbol{\mu}_N = \sum_A Z_A(\mathbf{R}_A - \mathbf{O}), \quad (5)$$

$$\boldsymbol{\mu}_{\text{tot}} = \boldsymbol{\mu}_e + \boldsymbol{\mu}_N. \quad (6)$$

For the finite molecule, $\mathbf{J}$ denotes total charge current, not a volume-normalized current density.

2.3 Time grid and sampling location

For a fixed step, integer index is authoritative:

$$t_n = t_0 + n\Delta t, \quad n = 0, \dots, N. \quad (7)$$

There are N propagation intervals and $N + 1$ endpoint states. Floating-point time is stored as a convenience. State, dipole, current, and sampled energy records are endpoint-centred. Work increments, point power, and Ward residuals associated with a propagation interval are midpoint- or interval-centred and carry their own step/time arrays.

3 Electronic structure and the AO equation of motion

3.1 Field-free Kohn–Sham model

The all-electron molecular reference is a converged restricted Kohn–Sham calculation. For a pure LDA/GGA functional its field-free Fock/Kohn–Sham matrix is written

$$H_0[P] = T + V_{\text{nuc}} + J[P] + V_{\text{xc}}[P]. \quad (8)$$

The corresponding field-free energy decomposition is

$$T_s[P] = \text{Re Tr}(PT), \quad E_{\text{eN}}[P] = \text{Re Tr}(PV_{\text{nuc}}), \quad (9)$$

$$E_{\text{H}}[P] = \frac{1}{2} \text{Re Tr}(PJ[P]), \quad E_{\text{xc}}[P] = E_{\text{xc}}[n_P], \quad (10)$$

$$E_{\text{NN}} = \sum_{A < B} \frac{Z_A Z_B}{R_{AB}}, \quad U_0[P] = T_s + E_{\text{eN}} + E_{\text{H}} + E_{\text{xc}} + E_{\text{NN}}. \quad (11)$$

Real-time propagation is used within the Runge–Gross density–potential framework[1]; PBE is a representative pure GGA in the validated functional class[5]. The implementation uses PySCF for the RKS reference, numerical grid, effective potential, and functional energy[6]. The first release stores combined exchange–correlation energy; an exchange/correlation split is not a required observable.

3.2 Connection in a time-dependent nonorthogonal basis

Projection of the time-dependent Kohn–Sham equation into an evolving basis gives

$$i\hbar \left(S\dot{C} + \omega C \right) = H_{\text{eom}} C, \quad \omega_{\mu\nu} = \langle \phi_\mu | \dot{\phi}_\nu \rangle. \quad (12)$$

The geometry is not an optional correction. It imposes metric compatibility

$$\dot{S} = \omega + \omega^\dagger, \quad (13)$$

which guarantees preservation of the physical overlap norm under the exact equation. Artacho and O’Regan discuss why an evolving Hilbert space is geometrically different from a standard fixed-coordinate matrix ODE[4].

For implementation, define

$$\mathcal{A} = S^{-1} \left[-\omega - \frac{i}{\hbar} H_{\text{eom}} \right], \quad \dot{C} = \mathcal{A}C, \quad \dot{P} = \mathcal{A}P + P\mathcal{A}^\dagger. \quad (14)$$

Knowledge of $S(t)$ and $\omega(t)$ at arbitrary times removes any need to integrate a separate metric equation. It does not remove the need to transport the electronic state consistently between two different metrics, nor the need to solve the nonlinear midpoint Hamiltonian self-consistently.

Status — Normative Aion 0.2. Every formulation returns the explicit triple $(S, H_{\text{eom}}, \omega)$. The older implicit split between an intrinsic Hamiltonian and a residual connection is not a public contract.

4 Electromagnetic sources and gauge data

4.1 Potential-first representation

The minimal-coupling one-particle operator is

$$h(t) = \frac{1}{2m} (\mathbf{p} - q\mathbf{a}(\mathbf{r}, t))^2 + q\phi(\mathbf{r}, t) + v_0, \quad (15)$$

where $\mathbf{p} = -i\hbar\nabla$. The electric field is

$$\mathbf{E} = -\partial_t \mathbf{a} - \nabla\phi. \quad (16)$$

A gauge function χ acts as

$$\mathbf{a}' = \mathbf{a} + \nabla\chi, \quad \phi' = \phi - \dot{\chi}, \quad \psi' = \exp(iq\chi/\hbar)\psi. \quad (17)$$

The reusable source layer describes the physical field separately from a chosen gauge representation. It is potential-first and evaluable at arbitrary times, including every SCEM midpoint. A field-first convenience input is compiled to consistent potentials before propagation; $\mathbf{a}$ is never constructed by incremental numerical integration inside the electronic hot loop. The compiled uniform specialization retains $\mathbf{a}$, $\dot{\mathbf{a}}$, and $\ddot{\mathbf{a}}$, and therefore $\mathbf{E} = -\dot{\mathbf{a}}$ and $\dot{\mathbf{E}} = -\ddot{\mathbf{a}}$, at every endpoint and midpoint. This supports analytic matter- and generator-energy derivatives without finite differences of the source.

4.2 Projected node/link sample

For the covariant formulation, associate AO μ with an atomic anchor $a(\mu)$. The compact projected source sample is

$$\left\{ \phi_a(t), L_{ab}(t), \dot{L}_{ab}(t) \right\}, \quad L_{ab} = \int_{\mathbf{R}_b}^{\mathbf{R}_a} \mathbf{a} \cdot d\boldsymbol{\ell}, \quad a < b. \quad (18)$$

The pair electromotive potential is

$$V_{ab} = -\dot{L}_{ab} - \phi_a + \phi_b. \quad (19)$$

Links and EMFs are oriented $b \rightarrow a$, while $I_{ab} > 0$ denotes charge flow $a \rightarrow b$. With the corresponding signed incidence matrix,

$$\dot{Q}_a + \sum_b I_{ab} = 0, \quad (20)$$

$$\mathbf{J}_{\text{graph}} = \sum_{a < b} I_{ab} (\mathbf{R}_b - \mathbf{R}_a), \quad (21)$$

$$\mathcal{P}_{\text{source}} = - \sum_{a < b} I_{ab} V_{ab}. \quad (22)$$

Under Eq. (17), $L'_{ab} = L_{ab} + \chi_a - \chi_b$, so Eq. (19) is invariant.

Multiple physical providers are added at the primitive potential/link level before projection or Wilson dressing. A temporal gate $g(t)$ acts on potentials,

$$\phi^g = g\phi, \quad L^g = gL, \quad \dot{L}^g = \dot{g}L + g\dot{L}. \quad (23)$$

Omitting the product-rule term would omit a real switching field.

4.3 Finite pulses and discrete kicks

A compactly supported resonant pulse may be defined in the vector potential:

$$\mathbf{a}(t) = a_0 \sin^2\left(\frac{\pi t}{T}\right) \sin(\Omega t + \varphi) \hat{\mathbf{e}}, \quad 0 \leq t \leq T, \quad (24)$$

with T an exact integer number of carrier cycles and $\mathbf{E} = -\dot{\mathbf{a}}$. The amplitude is normalized to the requested peak electric field. Equation (24) has $\mathbf{a}(0) = \mathbf{a}(T) = 0$ and zero net impulse. This is a property of this pulse family, not a restriction on the physical use of pulses.

If a maximum step is requested, choose

$$N = \lceil T/\Delta t_{\text{max}} \rceil, \quad \Delta t = T/N \quad (25)$$

so the boundary is exactly a grid point. General pulses may carry nonzero impulse. A simulation or analysis workflow may request a zero-impulse constraint when its scientific purpose requires one; the pulse abstraction itself imposes no such policy.

A delta kick is a uniquely identified discrete event, not a narrow sampled pulse. Events lie exactly on grid boundaries and are stored with pre- and post-event records at the same physical time, distinguished by side and sequence. The exact map is formulation-owned:

- for $\boldsymbol{\kappa} = \int \mathbf{E} dt$, bare LG applies the exact metric-unitary projected event $\exp[-iS^{-1}H_{\text{int}}/\hbar]$, with $H_{\text{int}} = -q\boldsymbol{\kappa} \cdot (\mathbf{r} - \mathbf{O}S)$; for an electron this has the continuum sign $\exp(-i\boldsymbol{\kappa} \cdot \mathbf{r})$;
- bare VG leaves coefficients continuous and makes the discontinuous change $\mathbf{a}(0^+) = -\boldsymbol{\kappa}$ in the adopted electronic sign convention;
- P0+E1 applies its covariant gauge/state event map.

Applied event identifiers are checkpointed, making a restarted event idempotent.

5 Bare length-gauge formulation

5.1 Equation of motion

For a uniform field in length gauge, $\mathbf{a} = 0$ and $\phi = -\mathbf{E} \cdot (\mathbf{r} - \mathbf{O})$. In a fixed AO basis,

$$S = S_0, \quad \omega = 0, \quad H_{\text{LG}}[P, t] = H_0[P] - q\mathbf{E}(t) \cdot (\mathbf{r} - \mathbf{O}S). \quad (26)$$

5.2 Dipole and currents

The primary current is the analytic time derivative of the length-gauge dipole:

$$\mu_{e,\alpha} = q \operatorname{Re} \operatorname{Tr}[PR_\alpha], \quad R_\alpha = r_\alpha - O_\alpha S, \quad (27)$$

$$\begin{aligned} J_{\text{LG},\alpha} &= q \operatorname{Re} \operatorname{Tr}[\dot{P}R_\alpha] \\ &= \Re \left\{ \frac{iq}{\hbar} \operatorname{Tr} \left[P \left(H_{\text{LG}} S^{-1} R_\alpha - R_\alpha S^{-1} H_{\text{LG}} \right) \right] \right\}. \end{aligned} \quad (28)$$

The second line follows from Eq. (14) and is not a finite difference.

For comparison, a local all-electron ambient mechanical diagnostic is

$$J_{\text{para},\alpha} = \frac{q}{m} \operatorname{Re} \operatorname{Tr}(Pp_\alpha), \quad J_{\text{dia},\alpha} = 0, \quad \mathbf{J}_{\text{mech}} = \mathbf{J}_{\text{para}}. \quad (29)$$

The analytic dipole current is the primary LG observable. In a finite basis, Eq. (28) and Eq. (29) need not coincide; their difference measures finite-subspace failure of continuum commutator identities.

5.3 Energy

The length-gauge matter energy excludes the source coupling:

$$U_{\text{matter}}^{\text{LG}} = U_0[P]. \quad (30)$$

The electronic and fixed-nuclear scalar couplings and generator energy are

$$U_{\text{EM},e}^{\text{LG}} = -\mathbf{E} \cdot \boldsymbol{\mu}_e, \quad U_{\text{EM},N}^{\text{LG}} = -\mathbf{E} \cdot \boldsymbol{\mu}_N, \quad (31)$$

$$E_{\text{generator}}^{\text{LG}} = U_{\text{matter}}^{\text{LG}} + U_{\text{EM},e}^{\text{LG}} + U_{\text{EM},N}^{\text{LG}}. \quad (32)$$

For fixed nuclei,

$$\frac{dU_{\text{matter}}^{\text{LG}}}{dt} = \mathbf{E} \cdot \mathbf{J}_{\text{LG}}, \quad (33)$$

$$\frac{dE_{\text{generator}}^{\text{LG}}}{dt} = -\dot{\mathbf{E}} \cdot \boldsymbol{\mu}_{\text{tot}}. \quad (34)$$

These are different valid energies. The phrase *total energy* is therefore insufficient without the qualifier **matter** or **generator**. When the field vanishes at both selected endpoints, the final deposited matter energy and final generator-energy gain agree.

6 Bare velocity-gauge formulation

6.1 Hamiltonian

For $\phi = 0$ and spatially uniform $\mathbf{a}(t)$, expansion of Eq. (15) gives the AO matrix

$$H_{\text{VG}}[P, t] = H_0[P] - \frac{q}{m} \mathbf{a}(t) \cdot \mathbf{p} + \frac{q^2}{2m} |\mathbf{a}(t)|^2 S. \quad (35)$$

This is the local all-electron velocity-gauge form used in the NAO RT-TDDFT treatment of Pemmaraju *et al.*[3]. For a nonlocal ionic potential, an additional gauge-transformed nonlocal operator and corresponding current correction are required; they are outside scope.

With PySCF derivative overlaps, the implementation fixes and tests the canonical-momentum convention

$$p_\alpha = i\hbar \langle \nabla_\alpha \phi_\mu | \phi_\nu \rangle, \quad (36)$$

including its Hermiticity and sign.

6.2 Mechanical current

The bare VG primary current is the mechanical current,

$$J_{\text{para}, \alpha} = \frac{q}{m} \text{Re Tr}(P p_\alpha), \quad (37)$$

$$J_{\text{dia}, \alpha} = -\frac{q^2}{m} N a_\alpha, \quad (38)$$

$$J_{\text{mech}, \alpha} = J_{\text{para}, \alpha} + J_{\text{dia}, \alpha}. \quad (39)$$

The two pieces must be available separately. Plotting only $q \text{Re Tr}(P p_\alpha)$ as the VG physical current omits the diamagnetic piece.

6.3 Mechanical kinetic and matter energy

The mechanical kinetic energy is

$$\begin{aligned} T_{\text{mech}}[P, t] &= T_{\text{can}}[P] + T_{\text{linear}}[P, t] + T_{\text{dia}}[P, t], \\ T_{\text{can}} &= \text{Re Tr}(PT), \\ T_{\text{linear}} &= -\frac{q}{m} \mathbf{a} \cdot \text{Re Tr}(P \mathbf{p}), \\ T_{\text{dia}} &= \frac{q^2}{2m} N |\mathbf{a}|^2. \end{aligned} \quad (40)$$

The VG matter energy is

$$U_{\text{matter}}^{\text{VG}} = T_{\text{mech}} + E_{\text{eN}} + E_{\text{H}} + E_{\text{xc}} + E_{\text{NN}}. \quad (41)$$

In the pure VG representation it is also the electronic generator energy for the prescribed field, and

$$\frac{dU_{\text{matter}}^{\text{VG}}}{dt} = \mathbf{E} \cdot \mathbf{J}_{\text{mech}}. \quad (42)$$

An energy that retains only T_{can} contains a large, apparently anomalous oscillatory error during the pulse. Both the linear $\mathbf{a} \cdot \mathbf{p}$ and quadratic a^2 contributions in Eq. (40) are compulsory.

7 Covariant P0 formulation

7.1 Wilson-dressed overlap

Let each AO carry an anchor position $\mathbf{R}_{a(\mu)}$. With the link oriented from the anchor of ν to that of μ , define

$$\Theta_{\mu\nu}(t) = \exp\left[\frac{iq}{\hbar}L_{a(\mu)a(\nu)}(t)\right], \quad S_{\mu\nu}(t) = \Theta_{\mu\nu}(t)S_{\mu\nu}^0. \quad (43)$$

The inverse-dressed density passed to the field-free electronic-structure model is

$$P^0 = \Theta^* \odot P, \quad (44)$$

where $\odot$ is elementwise multiplication. If $H_0[P^0]$ is constructed by PySCF, the P0 equation-of-motion matrix is obtained by the forward dressing

$$H_{P0}[P, t] = \Theta \odot H_0[P^0]. \quad (45)$$

7.2 Projected connection

Introduce the diagonal site scalar-potential connection

$$\sigma_{\mu\nu} = \delta_{\mu\nu} \frac{iq}{\hbar} \phi_{a(\mu)}. \quad (46)$$

The site-covariant overlap derivative and P0 connection are

$$D_{\text{site}}S = \dot{S} + \sigma S - S\sigma, \quad (47)$$

$$\omega_{P0} = S\sigma + \frac{1}{2}D_{\text{site}}S. \quad (48)$$

Because $\sigma^\dagger = -\sigma$ for a real scalar potential, Eq. (48) satisfies $\omega_{P0} + \omega_{P0}^\dagger = \dot{S}$ exactly.

The P0 equation is consequently

$$i\hbar(S\dot{C} + \omega_{P0}C) = H_{P0}[P, t]C. \quad (49)$$

The scalar-potential and link time dependence is geometrical: it is carried by the projected connection and metric, not added again to Eq. (45).

7.3 Site charge and graph current

Let M_a be the fixed AO selector/projector associated with anchor a . The symmetrized source charge is

$$Q_a = \frac{q}{2} \text{Re Tr}[P\{M_a, S\}]. \quad (50)$$

For each stored pair, let S_{ab} denote its oriented overlap contribution and $H_{\text{source},ba}$ the corresponding explicit source block. Define

$$K_{\text{dyn}} = H_{\text{dyn}} - \frac{i\hbar}{2}D_{\text{site}}S. \quad (51)$$

Here H_{dyn} is the Hermitian matrix in the equivalent coefficient-velocity equation after the non-metric connection terms have been moved to its right-hand side. Thus $H_{\text{dyn}} = H_{P0}$ for P0 and $H_{\text{dyn}} = H_{P0} + V_{E1}$ for P0+E1. This algebraic definition does not alter the normative connection split in Eq. (57). The common pair-current kernel used by the covariant model is

$$\begin{aligned} I_{ab} = & \frac{q}{\hbar} \text{Im Tr}\left[P\left(S_{ab}S^{-1}K_{\text{dyn}} + K_{\text{dyn}}^\dagger S^{-1}S_{ab}\right)\right] \\ & + \frac{2q}{\hbar} \text{Im Tr}(PH_{\text{source},ba}). \end{aligned} \quad (52)$$

At P0, $H_{\text{source}} = H_{\text{dyn}} = H_{\text{P0}}$, and one pair array provides both charge continuity and source response. At P0+E1 two distinct contractions of Eq. (52) are required. The full-dynamical array I_{ab}^{cont} uses $H_{\text{source}} = H_{\text{dyn}} = H_{\text{P0}} + V_{\text{E1}}$ and satisfies the site-charge continuity identity. The action-split source array $I_{ab}^{(\text{P0|E1})}$ uses the same full K_{dyn} , but sets $H_{\text{source}} = H_{\text{P0}}$; its Cartesian graph current is the P0 component to which Eq. (60) is added. Conflating the two arrays either breaks the site-continuity diagnostic or double counts the E1 phase response. They are therefore stored separately as `pair_currents_continuity` and `pair_currents_p0_source`.

8 P0+E1 correction

8.1 Central dipoles and variational connection

For the AO pair (μ, ν) , define the anchor-pair centre

$$\mathbf{R}_{\mu\nu} = \frac{1}{2} (\mathbf{R}_{a(\mu)} + \mathbf{R}_{a(\nu)}) \quad (53)$$

and field-free central dipole matrices

$$d_{\mu\nu}^{\alpha} = q \left[r_{\mu\nu}^{\alpha} - R_{\mu\nu}^{\alpha} S_{\mu\nu}^0 \right]. \quad (54)$$

Their dressed Hermitian form and E1 variational potential are

$$D_{\alpha}(t) = \text{Herm}[\Theta(t) \odot d^{\alpha}], \quad \text{Herm } X = \frac{1}{2}(X + X^{\dagger}), \quad (55)$$

$$V_{\text{E1}}[P, t] = - \sum_{\alpha} E_{\alpha}(t) D_{\alpha}(t). \quad (56)$$

The defining physics split is

$$\boxed{H_{\text{eom}} = H_{\text{P0}}, \quad \omega_{\text{P0+E1}} = \omega_{\text{P0}} + \frac{i}{\hbar} V_{\text{E1}}.} \quad (57)$$

Putting V_{E1} in both H_{eom} and ω double counts it. Equation (57) expresses the projection of the ambient electromagnetic connection into the P0+E1 equivariant subspace.

8.2 Covariant dipole

Writing $N_a = Q_a/q$, the electronic covariant dipole is

$$\mu_{\text{P0+E1}, \alpha} = q \sum_a N_a R_{a\alpha} + \text{Re Tr}(PD_{\alpha}). \quad (58)$$

The first term transports site charge between anchors; the second restores intra-/inter-site polarization retained at E1.

8.3 E1 residual and total source current

The reduced-vector-potential derivative of the dressed central dipole is

$$D_{\alpha; \beta} = \frac{\partial D_{\alpha}}{\partial a_{\beta}} = \text{Herm} \left[\frac{iq}{\hbar} (R_{a(\mu), \beta} - R_{a(\nu), \beta}) \Theta_{\mu\nu} d_{\mu\nu}^{\alpha} \right]. \quad (59)$$

The E1 residual source current is

$$J_{\text{E1}, \beta} = \frac{d}{dt} \text{Re Tr}(PD_{\beta}) + \sum_{\alpha} E_{\alpha} \text{Re Tr}(PD_{\alpha; \beta}). \quad (60)$$

Its first term is the intrinsic polarization rate and its second term is the phase response. Both pieces are recorded.

The total P0+E1 variational source current is

$$\mathbf{J}_{\text{src}} = \sum_{a < b} I_{ab}^{(\text{P0|E1})} (\mathbf{R}_b - \mathbf{R}_a) + \mathbf{J}_{\text{E1}}. \quad (61)$$

In Eq. (52), the coefficient/dynamical part sees the full P0+E1 equation, while the explicit source-Hamiltonian block is the base P0 source contribution. This separation prevents E1 double counting.

The defining checks are

$$\dot{Q}_a + \sum_b I_{ab}^{\text{cont}} = 0, \quad (62)$$

$$\mathbf{J}_{\text{src}} = \frac{d\boldsymbol{\mu}_{\text{P0+E1}}}{dt}, \quad (63)$$

$$\frac{dU_{\text{cov}}}{dt} = \mathbf{E} \cdot \mathbf{J}_{\text{src}}. \quad (64)$$

Status — Implemented and validated in WP3. The active 0.2 tree implements the P0/P0+E1 tensors, connection split, site charges, both pair-current contractions, and analytic E1-current pieces as backend-resident reusable kernels. Synthetic, H2, LiH, and physical-GPU tests check metric compatibility, continuity, variational source response, current-dipole agreement, gauge covariance, and the E1-once EOM split.

9 Mechanical-current diagnostics

9.1 Generic local contraction

For an arbitrary density and prescribed local vector potential, define AO matrices

$$A_{\alpha,\mu\nu}^{\text{op}}(t) = \langle \phi_\mu | a_\alpha(\mathbf{r}, t) | \phi_\nu \rangle. \quad (65)$$

The reusable low-level diagnostic is

$$J_{\text{para},\alpha} = \frac{q}{m} \text{Re Tr}(P p_\alpha), \quad (66)$$

$$J_{\text{dia},\alpha} = -\frac{q^2}{m} \text{Re Tr}(P A_\alpha^{\text{op}}), \quad (67)$$

$$\mathbf{J}_{\text{mech}} = \mathbf{J}_{\text{para}} + \mathbf{J}_{\text{dia}}. \quad (68)$$

For a uniform field, $A_\alpha^{\text{op}} = a_\alpha S$, giving Eq. (38). The density need only be finite, Hermitian, and shape-compatible; idempotency, positivity, and electron count are optional diagnostics rather than hidden preconditions.

9.2 Covariant ambient projection

In the uniform covariant representation the canonical momentum and diamagnetic operator are

$$p_\alpha^{\text{can,cov}} = \Theta \odot (p_\alpha^0 + q a_\alpha S^0), \quad (69)$$

$$p_\alpha^{\text{dia,cov}} = -q a_\alpha S(t). \quad (70)$$

Their explicit a_α terms cancel, so the total ambient-projected mechanical operator reduces to

$$p_\alpha^{\text{mech,cov}} = \Theta \odot p_\alpha^0. \quad (71)$$

The resulting observable is named `ambient_projected_mechanical_current`. It is a useful common diagnostic, but it is not the P0+E1 variational source current in Eq. (61). The two must never share a generic stored `current` label.

10 Covariant energy and universal energy ledger

10.1 Covariant matter energy

The gauge-invariant P0/P0+E1 matter energy is evaluated from the inverse-dressed density in Eq. (44):

$$U_{\text{cov}}[P, t] = T_s[P^0] + E_{\text{eN}}[P^0] + E_{\text{H}}[P^0] + E_{\text{xc}}[P^0] + E_{\text{NN}}. \quad (72)$$

The E1 source-coupling expectation

$$U_{\text{E1,coupling}} = \text{Re Tr}(PV_{\text{E1}}) \quad (73)$$

is stored separately and is not included in Eq. (72). This is necessary for the Ward identity Eq. (64). Equivalent dressed-frame contractions are valuable independent validation identities, not alternative user-facing definitions.

The representation-dependent instantaneous generator energy is stored as a different qualified quantity,

$$E_{\text{generator}}^{\text{cov}} = U_{\text{cov}} + \sum_a \phi_a Q_a + \sum_a Z_a \phi_a + U_{\text{E1,coupling}}, \quad (74)$$

where the final term is absent for P0. Equation (74) records the explicit source coupling appearing in the chosen representation; it is not used as the absorbed energy. Absorption and the covariant Ward test use the gauge-invariant matter energy in Eq. (72) and the formulation-owned variational source current.

10.2 Qualified totals

The energy observable records, when meaningful for a formulation:

1. canonical/paramagnetic kinetic, vector-potential-linear kinetic, diamagnetic kinetic, and their mechanical sum;
2. electron–nuclear, Hartree, combined exchange–correlation, and nuclear–nuclear energies;
3. electronic scalar EM coupling, fixed-nuclear scalar EM coupling, and their sum;
4. `energy_matter_total`;
5. `energy_generator_total`;
6. absorbed energy and accumulated source work;
7. analytic energy rate, source power, and Ward residuals.

The conventional DFT word *total* is retained only with a qualifier. There is no public `total_energy`, because Eqs. (33), (34), (42), and (64) show that different totals answer different physical questions during a pulse.

For every formulation,

$$\Delta U_{\text{abs}}(t) = U_{\text{matter}}(t) - U_{\text{matter}}(t_0) \quad (75)$$

is compared to independently accumulated source work. Fixed nuclei have no current and do not absorb power; their scalar coupling is nevertheless stored to establish origin cancellation in a neutral system.

10.3 Independent analytic energy derivative

For the pure-functional DFT energy, functional differentiation gives schematically

$$\frac{dU}{dt} = \text{Re Tr} [F_{\text{energy}}[P] \dot{P}] + \left. \frac{\partial U}{\partial t} \right|_P, \quad (76)$$

where $\dot{P}$ is obtained analytically from Eq. (14). This route is independent of the $\mathbf{E} \cdot \mathbf{J}$ contraction and therefore provides a non-tautological Ward test. Finite differences of sampled energies are only secondary convergence diagnostics.

At every accepted propagation interval, source work is advanced from the converged midpoint:

$$W_{n+1} = W_n + \Delta t \mathcal{P}_{\text{source}}(t_{n+1/2}). \quad (77)$$

The midpoint state and source already exist, so no additional Fock build is required. Work is accumulated every interval even when energy is not sampled there. A robust normalized Ward residual reports both absolute and relative scales and should decrease quadratically with Δt in the asymptotic regime.

Status — Implemented and validated in WP3. The qualified ledger, complete bare-VG kinetic decomposition, covariant energy, source couplings, analytic DFT functional derivative, source power, Ward residual, and analytic length-gauge generator rate are reusable backend-resident implementations. Energy remains independently scheduled and is not contracted at every step by default.

11 Self-consistent exponential midpoint propagation

11.1 Why the connection matters numerically

For a fixed linear ODE, exponentiating a midpoint generator is routine. Here three facts interact:

1. the density determines the Kohn–Sham Hamiltonian;
2. the coordinate metric differs at the beginning and end of an interval;
3. the connection contains both an analytically integrable diagonal site piece and a residual projected piece that belongs in the dynamical generator.

An exponential that is unitary in one Euclidean or Löwdin-orthogonalized frame need not map the S_n norm to the S_{n+1} norm. Small defects can accumulate and feed back into the nonlinear density. The remedy is to respect parallel transport and cross-metric geometry explicitly, not to pretend that $S(t)$ is fixed.

11.2 Common nonlinear midpoint solve

Both production propagators use the same SCEM fixed point. Given an input midpoint density, build $(S_{1/2}, H_{\text{com},1/2}, \omega_{1/2})$, construct a trial propagator, obtain an output midpoint density, and iterate to consistency. The convergence measure is

$$r_P = \frac{\|S_{1/2}^{1/2}(P_{\text{out}} - P_{\text{in}})S_{1/2}^{1/2}\|_F}{\max(N_e, 1)}. \quad (78)$$

The initial defaults are $r_P \leq 10^{-10}$, at most 50 iterations, and minimum damping 0.1. Start with a full Hermitian density update and reduce mixing when the residual worsens. Record the iteration count, final density residual, and an independent Hamiltonian residual. Only r_P determines fixed-point acceptance; the Hamiltonian residual is a separately recorded diagnostic, not a hidden second convergence condition. A failed midpoint is never accepted; tolerances and Δt are never silently changed.

11.3 Fixed-metric transport

For bare LG/VG, $S = S_0$ and $\omega = 0$. The midpoint generator is

$$\mathcal{A}_{1/2} = -\frac{i}{\hbar} S^{-1} H_{1/2}, \quad (79)$$

which is S -skew-Hermitian. Its rational map can therefore act directly in the nonorthogonal coordinates while preserving the S norm algebraically. A Löwdin transform is not required.

11.4 Connection-aware site-parallel frame

Write the analytically known diagonal site connection as $\sigma(t)$. Its exact parallel transport over one interval is

$$G_\sigma(t_{n+1}, t_n) = \exp \left[- \int_{t_n}^{t_{n+1}} \sigma(t) dt \right]. \quad (80)$$

For the diagonal Abelian uniform-field case this is elementwise and depends only on the known source. With physical reduced vector potential $\mathbf{a}$, AO anchor $\mathbf{R}_{a(\mu)}$, and electromagnetic origin $\mathbf{O}$, the implemented diagonal is

$$[G_\sigma(t, t_n)]_{\mu\mu} = \exp \left[-\frac{iq}{\hbar} (\mathbf{R}_{a(\mu)} - \mathbf{O}) \cdot (\mathbf{a}(t) - \mathbf{a}(t_n)) \right] \quad (81)$$

in the covariant length representation and the identity in the covariant velocity representation. Pull the overlap, Hamiltonian, and residual connection into this site-parallel frame. At the self-consistent midpoint, the combined generator is

$$\bar{\mathcal{A}}_{1/2} = \bar{S}_{1/2}^{-1} \left[-\bar{\omega}_{\text{res}, 1/2} - \frac{i}{\hbar} \bar{H}_{1/2} \right]. \quad (82)$$

The E1 contribution in Eq. (57) is part of $\bar{\omega}_{\text{res}}$, not an auxiliary length-gauge Hamiltonian.

11.5 Rational exponential and metric correction

The production default is the Padé [2/2] map

$$R_{22}(Z) = \left(I - \frac{Z}{2} + \frac{Z^2}{12} \right)^{-1} \left(I + \frac{Z}{2} + \frac{Z^2}{12} \right), \quad Z = \Delta t \bar{\mathcal{A}}_{1/2}. \quad (83)$$

Cayley [1/1] is retained as a diagnostic. These are systematically improvable approximations to the frozen exponential[7], but the complete midpoint method remains globally second order unless the Hamiltonian/source quadrature and nonlinear construction are also raised in order.

After transforming the raw interval map back to AO coordinates, impose the cross-metric constraint by right Cholesky correction:

$$\begin{aligned} M &= U_{\text{raw}}^\dagger S_{n+1} U_{\text{raw}} = R_M^\dagger R_M, \\ S_n &= R_S^\dagger R_S, \\ Q &= R_M^{-1} R_S, \quad U = U_{\text{raw}} Q. \end{aligned} \quad (84)$$

Then $U^\dagger S_{n+1} U = S_n$ up to roundoff. Record the raw defect, corrected defect, and $\|Q - I\|$. This is a constraint correction to an already constructed physical transport; it is not propagation in an evolving symmetrically orthogonalized Löwdin basis.

Correction disablement exists only as an explicit diagnostic construction. In that mode the raw defect is exposed rather than silently accepted as a production state constraint.

Only roundoff-scale anti-Hermitian contamination may be symmetrized, under a separately recorded threshold. A larger defect is a hard failure.

Status — Implemented and validated in WP4. One occupied-orbital state and one density-fixed-point SCEM engine serve all four formulations. Bare LG/VG use the direct nonorthogonal fixed-metric map; P0/P0+E1 use exact site transport, the pulled combined Hamiltonian–connection generator, and right-Cholesky cross-metric correction. Cayley [1/1] and Padé [2/2] share one backend-neutral solve path. Scratch arrays are allocated once per simulation, accepted midpoint data are retained in memory for observers, and every algebraic constraint or nonlinear failure is explicit. Synthetic, H2, and LiH tests establish norm/metric preservation, gauge covariance, approximately second-order molecular convergence, CPU/GPU agreement, physical GPU residency, and absence of Aion bulk host transfers inside a step.

12 Observable and storage model

12.1 Typed observable calculators

An observable declares:

- the formulation capabilities and operator/source dependencies it needs;
- whether it is evaluated at endpoints, midpoints, or events;
- its independent integer-step schedule;
- units, tensor shape, definition ID, and decomposition labels;
- whether evaluation requires a Fock build or may reuse midpoint data.

Unsupported observables fail during simulation construction. Energy is not computed every step by default. Energy, dipole/current, step diagnostics, checkpoints, and dense-matrix snapshots have independent schedules, while initial and final requested observables are always captured. The lightweight **PowerLedger** is evaluated from the already converged midpoint state on every accepted interval; it does not contract the complete energy-component ledger or rebuild the Fock matrix.

12.2 Canonical artifacts

The canonical numerical format is versioned HDF5:

Artifact	Purpose
<code>reference.h5</code>	Immutable backend-neutral prepared electronic reference
<code>trajectory.h5</code>	Canonical observable and diagnostic trajectory
<code>em_source.h5</code>	Optional full-resolution/per-provider source history
<code>checkpoint_<step>.h5</code>	Immutable restart state at an accepted boundary
<code>status.json</code>	Small atomic status record for detached monitoring

Top-level trajectory groups are

```
/meta, /configuration, /reference, /time, /source,  
/observables, /diagnostics, /events, /restart.
```

Definition-specific observable paths are required; generic `/current`, `/dipole`, or `/energy` datasets are forbidden. Each stream owns step/time coordinates. Every dataset has unit and dimension attributes. Scalar streams are uncompressed; dense/checkpoint arrays are chunked on record boundaries with low-level gzip and shuffle.

Writers publish transactionally: write a `.partial` file, validate and flush it, set an internal completion marker, and atomically create the final no-overwrite link before removing the private name. Completed artifacts are immutable. Controlled failures may publish `trajectory.failed.h5` with `complete=false`; a hard crash leaves only the partial artifact.

12.3 Restart and lineage

A checkpoint contains occupied coefficients, their accepted density, occupations, global step/time, accumulated work, the initial matter-energy baseline, runtime vector-potential offset, and applied-event IDs. It contains no pickled PySCF or GPU object. Restart reconstructs the prepared reference from normalized input and verifies all fingerprints.

A resumed segment first reconstructs, writes, and authenticates the repeated boundary state n_r before advancing to $n_r + 1$. It records parent run ID, checkpoint hash, and global step offset. Analysis deduplicates the shared boundary only through this lineage. SIGINT/SIGTERM is handled at a safe accepted-step boundary; a half-converged midpoint is never checkpointed as a valid state.

12.4 Semantic identity and provenance

A deterministic simulation ID is the SHA-256 hash of losslessly normalized scientific configuration, array dtypes/shapes/bytes, source and reference fingerprints. Paths, timestamps, host names, and output schedules are excluded. A unique run ID distinguishes execution attempts. Full software, Git, dependency, CPU/GPU, and device provenance is stored. A requested GPU run never falls back to CPU.

13 Spectroscopy

Casida linear response and real-time kick spectra are postprocessing workflows. Casida roots, oscillator strengths, and transition directions are stored as structured numerical data[2]. A resonant comparison selects a polarization-bright root by an explicit threshold or root choice and recomputes it for every basis; otherwise an on-resonance/off-resonance comparison would be physically ambiguous.

For a converged excitation n , let $\Omega_n > 0$ be its energy and $\mathbf{d}_n = \langle 0 | \hat{\boldsymbol{\mu}} | n \rangle$ its length-form transition dipole. The stored isotropic and polarization-resolved oscillator strengths are respectively

$$f_n = \frac{2}{3} \Omega_n |\mathbf{d}_n|^2, \quad f_n^{(e)} = 2 \Omega_n |\mathbf{e} \cdot \mathbf{d}_n|^2, \quad |\mathbf{e}| = 1. \quad (85)$$

The implementation calls PySCF’s restricted real Casida-TDDFT solver on a CPU mean field reconstructed from the authenticated prepared reference; it does not rerun SCF. Every returned root stores Ω_n , f_n , $\mathbf{d}_n$, the associated unit direction (zero for a dark transition), and its convergence flag. Resonance selection is either an explicit zero-based root or the lowest-energy root satisfying a stored threshold on $f_n^{(e)}$. The complete candidate-strength vector is retained, so the decision can be audited.

For an electric-field impulse $\boldsymbol{\kappa} = \int \mathbf{E}(t) dt$, define $\mathbf{e} = \boldsymbol{\kappa}/|\boldsymbol{\kappa}|$. One kick supplies the $\mathbf{e}$ -directed column of the response tensor. The canonical molecular response is computed from the formulation-owned electronic dipole,

$$\alpha_{\alpha\mathbf{e}}^{(\mu)}(\Omega) = \frac{\delta \tilde{\mu}_\alpha(\Omega)}{|\boldsymbol{\kappa}|}, \quad \delta \boldsymbol{\mu}(t) = \boldsymbol{\mu}(t) - \boldsymbol{\mu}_{\text{base}}. \quad (86)$$

For native Aion trajectories the default baseline is the exact pre-event dipole stored with the kick. A first-sample or zero baseline is possible only when explicitly requested. Linear kick spectroscopy accepts exactly one event and requires a complete dipole/current sample at every post-event endpoint; this prevents an unrecorded second perturbation or sparse sampling from being silently interpreted as a linear response.

The transform convention is

$$\tilde{f}(\Omega) = \int_0^T f(\tau) e^{-\eta\tau} e^{+i\Omega\tau} d\tau, \quad \tau = t - t_0, \quad (87)$$

where the finite record contains N samples separated by Δt , and $T = (N - 1)\Delta t$. Aion evaluates the continuous-time trapezoidal approximation

$$\tilde{f}(\Omega_k) \simeq \Delta t \left[\frac{1}{2}f_0 + \sum_{n=1}^{N-2} f_n e^{(-\eta + i\Omega_k)n\Delta t} + \frac{1}{2}f_{N-1} e^{(-\eta + i\Omega_k)T} \right] \quad (88)$$

using the positive-exponent inverse FFT. With $N_{\text{FFT}} = zN$, the stored positive grid is $\Omega_k = 2\pi k/(N_{\text{FFT}}\Delta t)$. The intrinsic finite-record resolution $2\pi/T$, native FFT spacing $2\pi/(N\Delta t)$, padded spacing $2\pi/(N_{\text{FFT}}\Delta t)$, and Nyquist energy $\pi/\Delta t$ are distinct stored quantities. Padding only interpolates; it cannot improve $2\pi/T$. The default finite-record window is rectangular. Optional exponential damping stores η in energy units and declares its isolated Lorentzian full width at half maximum 2η . The full complex transform, sign, normalization, time origin, quadrature, window, damping, maximum energy, and zero-padding factor are numerical data rather than plotting choices.

The directional absorption and oscillator-strength density are

$$A_e(\Omega) = \Omega \Im \left[\mathbf{e}^T \boldsymbol{\alpha}_e^{(\mu)}(\Omega) \right], \quad \rho_f^{(e)}(\Omega) = \frac{2}{\pi} A_e(\Omega). \quad (89)$$

Thus an isolated positive-frequency line integrates to the polarized strength in Eq. (85).

The independently evaluated formulation primary current obeys $\mathbf{J} = \dot{\boldsymbol{\mu}}$. Integration by parts on the same damped finite record gives the exact continuum identity

$$\tilde{\mathbf{J}}(\Omega) = \mathbf{B}(\Omega) + (\eta - i\Omega)\delta\tilde{\boldsymbol{\mu}}(\Omega), \quad (90)$$

$$\mathbf{B}(\Omega) = \delta\boldsymbol{\mu}(T)e^{(-\eta + i\Omega)T} - \delta\boldsymbol{\mu}(0). \quad (91)$$

The current-domain reconstruction is consequently

$$\boldsymbol{\alpha}_e^{(J)}(\Omega) = \frac{\tilde{\mathbf{J}}(\Omega) - \mathbf{B}(\Omega)}{|\boldsymbol{\kappa}|(\eta - i\Omega)}. \quad (92)$$

Aion stores both $\boldsymbol{\alpha}^{(J)}$ and the complex residual $\boldsymbol{\alpha}^{(J)} - \boldsymbol{\alpha}^{(\mu)}$. This is an independent physics check because the current is evaluated by the formulation, not obtained by finite-differencing the dipole. Retaining both endpoint terms is essential; omitting them produces a finite-record error even for exact time signals.

The expert API is the named package `aion.spectroscopy`. Casida and kick results are frozen validated numerical records. Their `aion.casida` and `aion.kick-spectrum` HDF5 artifacts are transactionally published, never overwritten, and content-authenticated. A spectrum identity includes its parent checksum, source/event/observable identities, transform choices, and numerical arrays but excludes the relocatable parent path. Knowledge of draft CSV/NPZ layouts is intentionally absent from Aion: the one-way importer in `CALCULATIONS/legacy/conversion` reads only numeric, atomic-unit data, records its complete field mapping and parent SHA-256, and creates a new immutable spectrum.

14 Reusable API and implementation boundaries

14.1 Static preparation and dynamic construction

The top-level Python workflow is intentionally small:

```
reference = prepare_reference(reference_config)
reference.save("reference.h5")

simulation = build_simulation(
```

```

    reference=reference,
    source=physical_source,
    formulation=formulation_config,
    propagation=propagation_config,
    observables=observable_config,
    output=output_config,
)
result = run(simulation)
resumed = resume(checkpoint_path, output=child_output)
trajectory = load_trajectory(path)

```

One prepared reference is reused by all formulations in a gauge comparison, preventing repeated SCF work and guaranteeing a common initial problem. References, configurations, operator bundles, sources, and formulation definitions are immutable. Backend caches and preallocated scratch arrays belong to a mutable `Workspace`; two simulations never share state or workspace.

14.2 Package domains

Package	Responsibility
<code>aion.config</code>	Immutable typed input, strict validation, normalization, units, and hashes.
<code>aion.backends</code>	Array protocol, CPU/GPU implementations, device residency, and workspaces.
<code>aion.electronic_structure</code>	PySCF RKS preparation, grids, reference reconstruction, core and derived operator bundles.
<code>aion.electromagnetism</code>	Physical sources, gauge representations, node/link samples, envelopes, compiled time series, and events.
<code>aion.formulations</code>	Bare LG/VG and P0/P0+E1 implementations of $(S, H_{\text{eom}}, \omega)$, formulation observables, and support checks.
<code>aion.propagation</code>	State protocols, common nonlinear SCEM, fixed-metric and connection-aware transport.
<code>aion.observables</code>	Typed dipole, current, mechanical-current, energy, power, continuity, and Ward calculators.
<code>aion.io</code>	HDF5 schemas, transactional writers, checkpoints, lineage, readers, and export.
<code>aion.spectroscopy</code>	Casida response, resonance selection, kick transforms, spectral observables, and immutable derived-result persistence.
<code>aion.workflows</code>	Preparation, simulation building, run/resume, and comparisons.

Dependencies are one-way: formulations consume references and sources; propagators consume the formulation protocol; observables inspect states and formulations; I/O receives records but never enters the physics. The CLI `aion prepare|run|resume|inspect|export` is a thin caller of the same typed API.

14.3 Prepared reference content

The immutable `reference.h5` contains normalized molecule/basis/ functional/grid input, converged coefficients and occupations, ground-state density, nuclear data, EM origin, exact

quadrature-grid coordinates and weights, and the core operator bundle

$$\{S, T, V_{\text{nuc}}, \mathbf{r}, \mathbf{p}, \text{nuclear data}, f\}.$$

It does not store AO values on the grid and never pickles live objects. Loading authenticates all content and dependency fingerprints. Constructing a fresh backend `Workspace` then reconstructs PySCF/GPU4PySCF and transfers the immutable arrays exactly once, without rerunning SCF. Atomic projectors/topology and P0/E1 tensors are independently versioned derived bundles with their own invalidation boundaries.

15 Present implementation versus the 0.2 target

Status — Current-tree assessment after WP5. The research draft was archived and verified on 8 September 2026. Its molecule-specific campaigns, runners, postprocessing, historical notes, and large results were moved with checksum manifests to the separate `CALCULATIONS` repository. WP1 then removed the active flat prototype modules and established the clean 0.2 package contracts. WP2 implements the common static electronic problem and prescribed electromagnetic input. WP3 implements the four formulation and instantaneous-observable layers after re-deriving and independently testing the archived equations. Draft equations and kernels remain available through the immutable archive and calculation repository, but none is silently reachable through the 0.2 API. WP4 adds the shared propagator and both transport geometries. WP5 adds the clean reusable runner, exact events, observers, transactional persistence, safe cancellation, authenticated restart, and operational CLI without reviving an archived molecule-specific runner. WP6 adds the named spectroscopy API, immutable Casida/kick-spectrum artifacts, and a one-way legacy conversion boundary outside the package.

The reusable layers implemented through WP6 are:

Current package	Implemented contract
<code>aion.config</code>	Frozen reference and simulation dataclasses, atomic-unit origin and fixed-grid types, discriminated formulations/sources/events, strict TOML resolution, and lossless canonical SHA-256 identities.
<code>aion.backends</code>	Strict NumPy and CuPy implementations of one array protocol, explicit host transfer, device-residency and wrong-device assertions, no GPU fallback, and a mutable per-simulation workspace for resident arrays, scratch, and live electronic-structure caches.
<code>aion.electronic_structure</code>	Validated CPU/GPU PySCF RKS preparation for the supported pure-LDA/GGA, all-electron, closed-shell domain. Immutable records contain the ground state, exact quadrature grid, core operators, nuclei, and independently fingerprinted AO-anchor/topology data. Loading authenticates portable arrays; workspace construction reconstructs PySCF/GPU4PySCF without another SCF calculation.
<code>aion.electromagnetism</code>	Potential-first uniform providers, additive composition, product-rule temporal gates, peak-field-normalized compact pulses, fixed-grid endpoint/midpoint compilation through $\ddot{\mathbf{a}}$, mutually consistent LG/VG/P0 node-link data, compiled-source persistence, and exact restart-idempotent kick definitions.

Current package	Implemented contract
<code>aion.io</code>	Independent semantic schemas and validators; transactional reference, source, trajectory, failure, checkpoint, Casida, and kick-spectrum publication; appendable independently sampled streams; compressed dense state; lazy readers; authenticated lineage stitching; CSV export; and atomic non-authoritative <code>status.json</code> .
<code>aion.observables</code>	Definition-specific immutable metadata and sampled records, plus typed, dependency-declared calculators with independent schedules and persistent formulation-qualified identities for dipoles, currents, energies, power, Ward, and continuity quantities.
<code>aion.formulations</code>	Explicit bare LG/VG and covariant P0/P0+E1 triples, Wilson geometry, inverse-dressed pure-DFT builds, E1 tensors, formulation-owned dipoles/source currents, mechanical diagnostics, graph continuity, source power, and qualified energy ledgers. All arrays remain resident on the selected NumPy/CuPy backend.
<code>aion.propagation</code>	Occupied-orbital state and diagnostics, shared damped-Picard SCEM fixed point, direct fixed-metric and analytic-connection transports, Cayley and Padé maps, right-Cholesky cross-metric correction, roundoff-only cleanup, hard constraint failures, and persistent backend-resident scratch.
<code>aion.spectroscopy</code>	Structured PySCF Casida-TDDFT roots, length-form and polarized oscillator strengths, explicit resonance selection, positive-frequency trapezoidal transforms, dipole/current kick response, resolution metadata, and immutable result I/O.
<code>aion.workflows</code> and <code>aion</code>	Typed <code>prepare_reference</code> , <code>load_reference</code> , <code>build_simulation</code> , <code>run</code> , <code>resume</code> , and <code>load_trajectory</code> workflows; exact runtime source events; one-run-per-process execution; independent-trajectory comparison manifests; and operational <code>prepare run resume inspect export</code> commands.

The reusable implementation through spectroscopy is complete. Legacy CSV/NPZ knowledge remains outside this package table because it belongs to the separate `CALCULATIONS/legacy/conversion` tree rather than the Aion runtime.

The physical-GPU WP2 gate caught a material scientific backend discrepancy. GPU4PySCF initially discarded the prebuilt PySCF quadrature grid and constructed a different point set. Energy agreement alone did not reveal the contract violation. Aion now installs the canonical stored coordinates and weights explicitly on the device before direct GPU SCF and before every live GPU reconstruction. H2 and LiH tests compare direct CPU/GPU preparation, ground-state densities, exact grids, reconstructed effective potentials, and the residency of every transferred reference and source array. The canonical momentum sign is additionally tested against an independent real-space ket-derivative quadrature, not only against libcint Hermiticity.

At present, some comparison scripts reconstruct missing VG linear and diamagnetic kinetic energies, calculate bare-LG momentum current, or assemble covariant mechanical diagnostics after propagation. Plotting scripts also encode observable definitions. In 0.2 these calculations belong to `aion.observables`; campaigns select and plot stored definitions but do not rederive

them.

16 Validation requirements

16.1 Hierarchy

The validation fixtures have distinct purposes:

Synthetic matrices

Exact signs, Hermiticity, metric compatibility, graph incidence/orientation, gauge covariance, rational-map algebra, and cross-metric transport.

H₂ Zero field, molecular symmetry, bare-gauge smoke tests, exact kicks, and compact timestep convergence.

LiH Nonzero E1 response, gauge/origin transformations, variational current, continuity, Ward energy, connection transport, restart, and CPU/GPU parity.

NH₃ Integrated release validation only: aug-cc-pVTZ, five driven cycles plus five field-free cycles, all four formulations, accepted step and a shorter half-step interval.

CO, ammonia/PNA production studies, and basis-convergence campaigns are external scientific calculations, not unit-test decisions. Existing draft trajectories are evidence, not golden truth.

16.2 Independent internal checks

The principal assertions are:

1. $P = P^\dagger$, electron number and occupied-state metric normalization;
2. Eq. (13) and cross-metric preservation;
3. explicit gauge-function transformations and EMF invariance;
4. translation of the EM origin, including electronic/nuclear cancellation for neutral systems;
5. Eq. (62) on every node and consistency of pair signs with Eq. (21);
6. directional derivatives of the variational source action versus Eq. (61);
7. analytic Eq. (63), independent of finite differences;
8. energy-component sums and analytic Eq. (76) versus source power;
9. approximately second-order reduction of trajectory and Ward errors under step halving;
10. uninterrupted versus restarted trajectory identity;
11. CPU/GPU parity for state, observables, energy components, invariants, and backend residency.

Full-array pinned trajectories are not the primary oracle. Selected numerical endpoints are retained only when they provide an independent protection.

16.3 GPU and performance policy

Every milestone that changes executable numerical physics must run its applicable tests on the physical GPU. The execution sandbox may hide the device; this is handled by requesting elevated execution, never by omitting GPU tests or substituting CPU-only tests. A requested GPU backend fails if unavailable.

The fast test suite uses up to eight workers with one BLAS thread each. Molecular CPU tests run serially with up to eight BLAS threads, and GPU tests run serially. Preparation, Fock construction, steady step time, memory, I/O, and host transfers are measured separately. An unexplained steady-step regression of roughly ten percent or more is flagged, but correctness and physics gates take priority.

17 Conclusions

The bare and covariant formulations solve related physical problems but do not share interchangeable current or energy definitions. Bare LG uses the analytic dipole derivative as its primary current; bare VG uses paramagnetic plus diamagnetic mechanical current and requires the complete mechanical kinetic energy; P0+E1 uses the variational source current implied by its projected connection. A common ambient-projected mechanical current is an important diagnostic, not a replacement for the covariant source current.

The difficult numerical feature of P0+E1 is not ignorance of the connection: the connection is analytically known from prescribed fields and static matrix elements. The issue is faithful finite-step transport between distinct metrics while the nonlinear Hamiltonian is determined at the midpoint. Exact site parallel transport, a combined residual generator, Padé midpoint propagation, and a right Cholesky cross-metric correction address that geometry without an evolving Löwdin frame.

The 0.2 architecture makes these physics definitions reusable and auditable. It separates immutable reference preparation, physical sources, formulations, propagation, observables, and storage; it records qualified energy/current definitions; and it treats conservation, Ward identities, convergence, restart, and CPU/GPU parity as release gates. The preservation and repository-separation gate WP0, contract gate WP1, common-reference/source gate WP2, instantaneous-physics gate WP3, shared-propagation gate WP4, and reusable execution/persistence gate WP5, and spectroscopy/legacy-boundary gate WP6 are complete. Integrated validation, profiling, and release remain separately gated in WP7.

Rebuilding this document

From the `aion/docs` directory, the PDF is rebuilt and all references are resolved with the single shell command

```
for pass in 1 2 3; do \  
  /usr/bin/pdflatex -interaction=nonstopmode \  
  -halt-on-error theory_and_implementation.tex; \  
done
```

The generated auxiliary, log, outline, and table-of-contents files are build artifacts; the tracked deliverables are the TeX source and PDF.

References

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